coming years will lead to the utmost resource utilisation by resource sharing. Instead of buying multiple systems for different operating systems, one can buy just one operating system (OS), set it up online on the cloud, and operate it liberally. This also helps in reducing the amount of e-waste generated.

***Virtualisation.*** Virtualisation has come from its root word virtual, which means 'not real.' The concept of virtualisation puts forth a mirage of resources in front of IT users. The concept of virtualisation in system memory brings in the utilisation of memories and showing them to be much larger than these actually are. The same concept is used in cloud computing and other similar technologies.

A virtual box installed on a computer system can do wonders these days. One can buy resources online from vendors and install a virtual box, irrespective of its compatibility with the system and thereby enjoy all sorts of services. Virtualisation indeed is one of the most important pillars of green computing.

## Green computing

To promote green computing concept, the following approaches are employed:

- Minimising energy consumption in an eco-friendly manner
- Safe disposal of e-waste
- Energy-efficient design of electronics and computer products
- Minimising waste during the manufacturing of computer hardware
- Using computers in power-saving modes when away
- Power management
- Re-use of electronics products

***Green electronics.*** As we all know, electronics forms the base of IT. Green electronics is a way of emphasising on disposing of hazardous substances that come from electronic waste or e-waste by using eco-friendly techniques. The e-waste is nothing but the waste generated by electronic products like computers, mobile phones, television sets, cables, wires, batteries, etc. E-waste consists of elements like As, Cd, Pb, and Hg and compounds like plastics, CFCs, PVCs, etc, that are hazardous.

The ewastes are usually disposed of by burning. The whole process is subject to a lot of emissions, which makes the environment poisonous and unfit, inviting unwanted acid rain, global warming, and other environmental concerns.

According to Green Electronics Council (GEC), the modern-world IT contributes a lot to e-waste due to invention and production of IT products made up of toxic substances. The EPEAT, or Electronic Product Environmental Assessment Tool, launched by GEC, ensures that only sustainable IT products are designed, manufactured, and purchased. GEC ensures that both manufacturers and large scale purchasers:

- Understand the problems faced due to e-waste
- Commit to address those challenges, and
- Act wisely to modify the operational and procurement processes with regards to the manufacturing of products by sticking to EPEAT standards

EPEAT is the leading global eco-label of the IT sector. National governments, including the United States and thousands of private and public institutional purchasers around the world, use EPEAT as part of their sustainable procurement decisions.

### Role of AI and IoT

- A good and efficient amalgamation of hardware and software used to set up AI and IoT systems comparatively reduces the amount of e-waste generation.
- Technologies like these can be exploited to build systems to monitor hardware.
- Using systems like these assist in reducing a lot of carbon footprints.
- AI and IoT systems can be set up with a perspective to reduce air pollution. AI governed systems like these can identify the source of pollution while incinerating e-waste with the help of smart sensors, so that remedial measures can be applied efficiently.

## Green startups

Many companies have come up recently with their focus on environmental protection. Some of these are:

***Attero Recycling.*** Started in 2008 in Noida, Attero Recycling is one of the few companies in the world working on proper e-waste management.

***Ather Energy.*** It is a Bengaluru based EV startup that manufactures its own e-scooters and offers charging infrastructure through the 'Ather Grid.'

***Yulu.*** It is a Bengaluru based startup that aims to prevent traffic congestion and air pollution by manufacturing a number of bicycles and lightweight electric two-wheelers governed by technologies like artificial intelligence (AI), machine learning (ML), and the Internet of Things (IoT).

***Ultraviolette.*** Ultraviolette has recently launched an electric motorcycle powered by modular battery technology, with a self-sustaining advanced and powerful electronic device.

## Recent findings and initiatives on green IT and electronics

Many researchers worldwide are investigating various areas in order to find some ways of altering architectures of these IT devices to bring in energy-efficient methods of computation, communication, or data transfer. Analio Fernandez and his fellow researchers at Cambridge University have tried to investigate the same by using nanotechnology and magnetisation. IT revolution has been possible due to the development of miniature integrated circuits (ICs) that are based on millions of transistors, which are, in turn, formed on silicon chips.

These light but powerful devices are reasons for high amounts of processing in any electronic device, but this leads to high power consumption as well. These silicon chips, which form a major part of computers, are nano-metric in size. The bits of information are stored in the form of an electrical charge in capacitors. Computation bits keep moving from memory to processors and again move back to memory after processing.

As we are wasting a lot of energy in continuous energy supply and movement of bits back and forth between memory and processor, current computing architectures need to be more efficient.

Hence, researchers have come up with the idea of using magnetic nanostructures for green computation. Special attention is given to the magnetic nanowires, which are several